# MATHEMATICAL FOUNDATIONS *of* QUANTUM MECHANICS

*New Edition*

JOHN VON NEUMANN

*Edited by* NICHOLAS A. WHEELER

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*John von Neumann talking shop with visiting scholars during afternoon tea at the Institute for Advanced Study. (Photo by Alfred Eisenstaedt / The LIFE Picture Collection / Getty Images.)*

# MATHEMATICAL FOUNDATIONS
OF
# QUANTUM MECHANICS

*by*

**John von Neumann**

NEW EDITION

*translated from the German by*

**ROBERT T. BEYER**

*edited by*

**Nicholas A. Wheeler**

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# TRANSLATOR'S PREFACE

This translation follows closely the text of the original German edition. The translated manuscript has been carefully revised by the author so that the ideas expressed in this volume are his rather than those of the translator, and any deviations from the original text are also due to the author.

The translator wishes to express his deep gratitude to Professor von Neumann for his very considerable efforts in the rendering of the ideas of the original volume into a translation which would convey the same meanings.

Robert T. Beyer

Providence, R. I.
December, 1949

# PREFACE TO THIS NEW EDITION

This book is the realization of my long-held intention to someday use the resources of TEX to produce a more easily read version of Robert T. Beyer's authorized English translation (Princeton University Press, 1955) of John von Neumann's classic *Mathematische Grundlagen der Quantenmechanik* (Springer, 1932).

I have endeavored to correct typographic errors that crept into the text and equations in the Beyer translation (a few of which were copied from the German original) and to adjust some of the transliterated sentences so that they read more easily and their meaning is easier to comprehend. But in all other respects I have adhered as closely as possible to the Beyer translation, which von Neumann reviewed and endorsed. Von Neumann wrote about his own extensive contributions to the Beyer translation in a letter (3 October 1949) addressed to Hayward Cirker, founding president of Dover Publications; that letter is reproduced on pages 91–92 of Miklós Rédei's *John von Neumann: Selected Letters* (2005).

The glorified typewriter (VariTyper, circa 1945) that was used to prepare the Beyer translation provided relatively few typographic resources, with the consequences that some of the equations are very awkwardly displayed and it is often not possible to know—except from context—the mathematical nature of the object to which a given symbol refers. Additionally, some of the symbols employed (for example, the German Gothic $\mathfrak{R}$,$\mathfrak{N}$ and certain Roman fonts) are difficult to distinguish in small type sizes. To avoid such difficulties, I have adopted the following invariable conventions:

- functions are denoted $f, g, h, \ldots$
- linear manifolds are denoted $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$, $\ldots$, $\mathcal{R}$, $\mathcal{S}$, $\ldots$, $\mathcal{M}$, $\mathcal{N}$, $\ldots$
- elements of linear manifolds are denoted $\phi, \psi, \ldots$
- physical quantities are denoted $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$, $\ldots$
- operators are denoted $\mathsf{A}$, $\mathsf{B}$, $\mathsf{C}$, $\ldots$, $\mathsf{E}$, $\ldots$, $\mathsf{P}$, $\mathsf{Q}$, $\ldots$, $\mathsf{U}$, $\mathsf{V}$
- matrices are denoted $\mathbb{A}, \mathbb{B}, \mathbb{C}, \ldots, \mathbb{P}, \mathbb{Q}, \ldots$
- 3-vectors are denoted $\boldsymbol{A}, \boldsymbol{x}, \boldsymbol{y}, \ldots$

Where von Neumann invariably writes $h/2\pi$ I employ Dirac's (now universal) symbol $\hbar$.

It is my impression that von Neumann's monograph—which is today, after more than eighty years, more frequently cited than ever before—is perhaps more frequently cited than actually read, and is my certain knowledge that it contains

treasures and provides valuable insights that are too seldom appreciated. It is my hope that this edition (to which I have attached a subject index, absent from previous editions) will facilitate—and perhaps inspire—the work of a new generation of physicists.

I thank especially Marina von Neumann Whitman, Freeman Dyson and Peter Renz, and also Jeffrey Bub, Auletta Gennaro, David Griffiths, David Mermin, Miklós Rédei, Robert Rohmer, Maximilian Schlosshauer, Silvan Schweber, Veeravalli Varadarajan, Thomas Wieting and Wojciech Zurek for encouragement in this undertaking. I am greatly indebted to William Bialek of the Princeton physics faculty for his initiative, and to Vickie Kearn for her expert editorial supervision. It is with special pleasure that I thank Linda Maddux, Reed College science librarian, for assistance in preparation of the list of works cited, Paul Forman and (especially) Harvey Leff for editorial advice and assistance. The project would never have come to completion but for the wise counsel and ever-tactful patience of my attorney, William Drew, of Elliott, Ostrander & Preston, P.C. And I am—as are we all—deeply indebted to Donald Knuth for his invention of TEX.

This volume was written in Plain TEX, as implemented by Textures, an application developed here in Portland by Blue Sky Research. Textures runs, however, only on OSX systems old enough to provide Classic Mode, so the work was done on a treasured antique of an iMac.

Nicholas Wheeler

Reed College
June, 2014

*By way of a*

# FOREWORD

*In 1933, John von Neumann [28 December 1903–8 February 1957] became (together with Albert Einstein and Oswald Veblin) a member of the founding faculty (and Kurt Gödel a visiting member) of the Institute for Advanced Study, where he remained for the remainder of his life. The distinguished author of the following commemorative essay was a member in 1949 and 1952–1954, and had then an opportunity to become personally acquainted with von Neumann. The essay was taken (with permission) from volume 64, number 3, part 2 (May 1958) pages 95–99 of the Bulletin of the American Mathematical Society, the 129-page entirety of which is given over to expert surveys of von Neumann's contributions to eight different subject areas.*

## VON NEUMANN'S CONTRIBUTIONS TO QUANTUM THEORY

LÉON VAN HOVE

That von Neumann has been "par excellence" the mathematician of quantum mechanics is as obvious to every physicist now as it was a quarter of a century ago. Quantum mechanics was very fortunate indeed to attract, in the very first years after its discovery in 1925, the interest of a mathematical genius of von Neumann's stature. As a result, the mathematical framework of the theory was developed and the formal aspects of its entirely novel rules of interpretation were analyzed by one single man in two years time (1927–1929). Conversely, one could almost say in reciprocity, quantum mechanics introduced von Neumann into a field of mathematical investigation, operator theory, in which he achieved some of his most prominent successes.

Von Neumann's major contributions to quantum mechanics are his development of the mathematical framework of the theory and his formal study of quantum statistics, quantum measuring processes and their interrelations. Whereas the latter study was essentially complete in 1927 (except for the quantum ergodic theorem of 1929) the work on the mathematical foundations of quantum mechanics came to its culmination in 1929 with the spectral theorem

for hypermaximal symmetric operators in Hilbert space. In the next two paragraphs we shall discuss these major contributions.

**The mathematical framework of quantum theory.** By the time von Neumann started his investigations on the formal framework of quantum mechanics this theory was known in two different mathematical formulations: the "matrix mechanics" of Heisenberg, Born and Jordan, and the "wave mechanics" of Schrödinger. The mathematical equivalence of these formulations had been established by Schrödinger, and they had both been embedded as special cases in a general formalism, often called the "transformation theory," developed by Dirac and Jordan. This formalism, however, was rather clumsy and it was hampered by its reliance upon ill-defined mathematical objects, the famous delta-functions of Dirac and their derivatives. Although von Neumann himself attempted at first, in collaboration with Hilbert and Nordheim [1], to edify the formalism along similar lines, he soon realized that a much more natural framework was provided by the abstract, axiomatic theory of Hilbert spaces and their linear operators [2]. This mathematical formulation of quantum mechanics, whereby states of the physical system are described by Hilbert space vectors and measurable quantities by Hermitian operators acting upon them, has been very successful indeed. Unchanged in its essentials it has survived the two great extensions which quantum theory was to undergo soon: the relativistic quantum mechanics of particles (Dirac equation) and the quantum theory of fields.

One might of course remark that Dirac's delta functions and their derivatives, although poorly defined at the time of their introduction, have been recognized since as bona fide mathematical entities in L. Schwartz' theory of distributions. This is quite true and moreover these functions have been used continually by physicists throughout the development of quantum theory, in particular in the last two decades for the study of scattering processes and of quantized fields. Delta functions have established themselves as the natural tool whenever operators with continuous spectra are to be considered. This does not affect in any way, however, the fact that the axiomatically defined separable Hilbert space is the suitable framework for the quantum-mechanical formalism as we know it today, and the recognition of this fact we owe to von Neumann.

An essential feature of the Hilbert space formulation of quantum theory is that the most important physical quantities (such as position, momentum or energy) are represented by unbounded Hermitian operators. Since the theoretical prediction of measurements makes essential use of the spectral resolution of the operators representing the physical quantities, von Neumann was, in his very first invetigation [2], faced with the problem of extending to the unbounded case the known spectral theory of bounded Hermitian operators. By 1929 he had brought this problem to a complete solution [3]. He introduced the all-important concept of hypermaximal symmetric operator, being the most general Hermitian operator with a spectral resolution. This work, the results of which were reached independently by M. H. Stone [4], was for von Neumann

the start of a long series of investigations on linear operators in Hilbert space.

Still another contribution of von Neumann to the mathematical foundation of quantum theory is worth mentioning here. He established the important theorem that (in the irreducible case and after a suitable reformulation) the canonical commutation rules $Q_j P_\ell - P_\ell Q_j = \hbar i \delta_{j\ell}$ determine the operators $Q_1, \ldots, Q_n, P_1, \ldots, P_n$ uniquely except for an arbitrary transformation [5]. Although rarely quoted as such, this theorem, which was already known to Dirac and Stone [6], is really fundamental for the understanding of many quantum-mechanical investigations where the theoretical analysis is exclusively based on the canonical commutation rules or in the equivalent field-theoretical form

$$A_j A_\ell^* - A_\ell^* A_j = \hbar \delta_{j\ell}, \qquad 2^{\frac{1}{2}} A_j = P_j - iQ_j$$

**Statistical aspects of quantum theory.** In the course of his formulation of quantum mechanics in terms of vectors and operators of Hilbert space von Neumann also gave in complete generality the basic statistical rule of interpretation of the theory. This rule concerns the result of the measurement of a given physical quantity on a system in a given quantum state and expresses its probability distribution by means of a simple and now completely familiar formula involving the vector representing the state and the spectral resolution of the operator which represents the physical quantity [2]. This rule, originally proposed by Born in 1926, was for von Neumann the starting point of a mathematical analysis of quantum mechanics in entirely probabilistic terms. The analysis, carried out in a paper of 1927 [7], introduced the concept of statistical matrix for the description of an ensemble of systems which are not necessarily all in the same quantum state. The statistical matrix (now often called $\rho$-matrix although von Neumann's notation was $U$) has become one of the major tools of quantum statistics and it is through this contribution that von Neumann's name became familiar to even the least mathematically minded physicists.

In the same paper von Neumann also investigates a problem which is still now the subject of much discussion; viz., the theoretical description of the quantum-mechanical measuring process and of the non-causal elements which it involves. Mathematically speaking von Neumann's study of this delicate question is quite elegant. It provides a clear-cut formal framework for the numerous investigations which were needed to clarify physically the all-important implications of quantum phenomena for the nature of physical measurements, the most important of which is Neils Bohr's concept of complementarity.

The results of the paper just discussed were immediately used by the author to lay the foundation for quantum thermodynamics [8]. He gave the quantum analogue

$$S = -k \operatorname{tr}(\rho \ln \rho), \qquad \rho \text{ denotes a statistical matrix}$$

of the well known classical formula for the entropy

$$S = -k \int f \ln f d\omega, \qquad f \text{ denotes a distribution function in phase space.}$$

He further wrote down the density matrix for a canonical ensemble at temperature $T$:

$$\rho = Z^{-1} \exp(-H/kT), \qquad Z = \mathrm{tr}\big[\exp(-H/kT)\big],$$

$H$ being the Hamiltonian operator. Two years later von Neumann came back to quantum thermodynamics with a contribution to a much more difficult problem: the formulation and proof of an ergodic theorem for quantum systems [9]. The basic principle of this work is to define quantum analogues of cells in phase space by considering sets of quantum states for which all macroscopic quantities have given values with a certain inaccuracy. One further considers the unitary transformation $u$ relating these quantum states to the eigenstates of the Hamiltonian. The ergodicity is then established for "almost every" value of the transformation $u$. Although the latter restriction is a rather unsatisfactory one from a physical standpont, one must consider von Neumann's ergodic theorem as one of the very few important contributions to a most difficult subject which even now is far from complete clarification.

Most of the work we have briefly reviewed has been republished by the author, in greatly expanded form, as a book which rapidly became and still is the standard work on the mathematical foundations of quantum mechanics [10], [and of which the present volume is an English translation]. Von Neumann devoted in his book considerable attention to a point which had not been discussed in the 1927 papers and which was later the subject of much controversy. It is the question of the possible existence of "hidden variables," the consideration of which would eliminate the non-causal element involved in the measurement process. Von Neumann could show that hidden parameters with this property cannot exist if the basic structure of quantum theory is retained. Although he mentioned the latter restriction explicitly,† his result is often quoted without due reference to it, a fact which sometimes gave rise to unjustified criticism in the many discussions devoted through the years to the possibility of an entirely deterministic reformulation of quantum theory.

**Other contributions.** As von Neumann's complete bibilography will reveal, he wrote quite a few other papers on quantum mechanics, often in collaboration with physicists, especially with Wigner. Most of these papers deal with technical matters and the importance of the major contributions discussed above is so eminent that, in comparison, the other papers' scope is modest. There is only one broad subject which we would like to mention here, because von Neumann, obviously giving it considerable thought, returned to it several times in 1934 and 1936 (in collaboration with Jordan, Wigner and Garrett Birkhoff). It is the question of the algebraic and logical structure of quantum mechanics, where the hope has existed to reach through abstract analysis possible generalizations of the accepted theory. Nobody knows whether such a hope is justified, but it is undoubtedly a natural one and it has appealed to many other people, giving one more example of the power and originality of von Neumann's thinking.

† See, *e.g.*, [10], p. 109, line 17 *et seq* [or below: p. 132, paragraph 3].

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**10.** ———, *Mathematische Grundlagen der Quantenmechanik*, Berlin, J. Springer (1932).

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*A* "von Neumann Days" Symposium *took place on 3–7 May 2010 at Brown University's Center for Computational Molcular Biology. Remarks prepared for presentation on that occasion by Freeman Dyson—who was a member of the Institute for Advanced Study in 1948–1949 and became there a tenured faculty colleague of von Neumann in 1953—were subsequently published as "A walk through Johnny von Neumann's garden" (Notices of the AMS* ***60****, 155–161 (2013)), from which the following excerpt has (with permission) been taken.*

## VON NEUMANN WORK RELATED TO QUANTUM THEORY

FREEMAN DYSON

During his Berlin years, Johnny made frequent visits to Göttingen, where Heisenberg had recently invented quantum mechanics and Hilbert was the presiding mathematician. Hilbert was intensely interested in quantum mechanics and encouraged collaboration between mathematicians and physicists. From the point of view of Hilbert, quantum mechanics was a mess. Heisenberg had no use for rigorous mathematics and no wish to learn it. Dirac made free use of his

famous delta-function, which was defined by a mathematical absurdity, being infinite at a single point and zero everywhere else. When Hilbert remarked to Dirac that the delta-function could lead to mathematical contradictions, Dirac replied, "Did I get into mathematical contradictions?" Dirac knew that his delta-function was a good tool for calculating quantum processes, and that was all that he needed. Twenty years later, Laurent Schwartz provided a rigorous basis for the delta-function and proved that Dirac was right. Meanwhile, Johnny worked with Hilbert and published a series of papers cleaning up the mess. In 1932 he published the book *Mathematical Foundations of Quantum Mechanics* [9], which occupies a substantial piece of his garden.

Johnny's book was the first exposition of quantum mechanics that made the theory mathematically respectable. The concepts were rigorously defined and the consequences rigorously deduced. Much of the work was original, especially the chapters on quantum statistics and the theory of measurement. I read the book in 1946 when I was still a pure mathematician but already intending to switch my attention to physics. I found it enormously helpful. It gave me what I needed, a mathematically precise statement of the theory, explaining the fine points that the physicists had been too sloppy to mention. From that book I learned most of what I know about quantum mechanics. But then, after I had made the transition to physics and had begun to read the current physics journals, I was surprised to discover that nobody in the physics journals ever referred to Johnny's book. So far as the physicists were concerned, Johnny did not exist. Of course, their ignorance of Johnny's work was partly a problem of language. The book was in German, and the first English translation was only published in 1955. But I think even if the book had been available in English, the physicists of the 1940s would not have found it interesting. That was a time when the culture of physics and the culture of mathematics were most widely separated. The culture of physics was dominated by people like Oppenheimer who made friends with poets and art historians but not with pure mathematicians. The culture of mathematics was dominated by the Borbaki cabal, which tried to expunge from mathematics everything that was not purely abstract. The gap between physics and mathematics was as wide as the gap between science and the humanities described by C. P. Snow in his famous lecture on the two cultures. Johnny was one of the very few people who were at home in all four cultures: in physics and mathematics, but also in science and the humanities.

The central concept in Johnny's version of quantum mechanics is the abstract Hilbert space. Hilbert space is the infinite-dimensional space in which quantum states are vectors and observable quantities are linear operators. Hilbert had defined and explored Hilbert space long before quantum mechanics made it useful. The unexpected usefulness of Hilbert space arises from the fact that the equations of quantum mechanics are exactly linear. The operators form a linear algebra, and the states can be arranged in multiplets defined by linear representations of the algebra. Johnny liked to formulate physical problems in abstract and general language, so he formulated quantum mechanics as a theory

of rings of linear operators in Hilbert space. A ring means a set of operators that can be added or subtracted or multiplied together but not divided. Any physical system obeying the rules of quantum mechanics can be described by a ring of operators. Johnny began studying rings of operators to find out how many different types of quantum systems could exist.

After Johnny published his quantum mechanics book, he continued for several years to develop the theory of rings of operators. The third volume of his collected works consists entirely of papers on rings of operators. He published seven long papers with a total of more than five hundred pages. I will not discuss these monumental papers this morning. They contain Johnny's deepest work as a pure mathematician. He proved that every ring of operators is a direct product of irreducible rings that he called factors. He discovered that there are five types of factor, of which only two were previously known. Each of the types has unique and unexpected properties. Exploring the ocean of rings of operators, he found new continents that he had no time to survey in detail. He left the study of the three new types of factor unfinished. He intended one day to publish a grand synthesis of his work on rings of operators. The grand synthesis remains an unwritten masterpiece, like the eighth symphony of Sibelius.

The quantum mechanics book is the last item on my list of flowers that Johnny published in German. It was published in 1932 when he was dividing his time equally between Berlin and Princeton. In the same year he began writing papers in English. One of his first papers to appear in English was "Proof of the quasi-ergodic hypothesis" [10], which he published in the *Proceedings of the National Academy of Sciences* to make sure that American mathematicians would read it. This paper solved an important problem in classical mechanics using the same concept of Hilbert space that he had used to solve problems in quantum mechanics. A classical dynamical system is said to be ergodic if after we put it into an initial state and then leave it alone for an infinite time, it comes arbitrarily close to any final state with probability independent of the initial state. Johnny proved that under certain clearly specified conditions, a system is ergodic if and only if there exist no constants of the motion. A constant of the motion means a quantity depending on the state of the system which does not change as the system moves forward in time. Johnny's theorem provides a firm mathematical basis for the assumptions that are customarily made by physicists using classical statistical mechanics. Translated into the sloppy language used by physicists, the theorem says that the time-average of any single trajectory of the system over a long time is equal to the statistical average of all trajectories. Even more sloppily, physicists say that time-averages are equal to ensemble averages, and we use the word ensemble to mean the set of all states of the system.

One of the American mathematicians who read Johnny's paper in the *Proceedings of the National Academy of Sciences* was Garrett Birkhoff. Garrett was the son of George Birkhoff, and both father and son were famous mathematicians. Garrett and Johnny became close friends, and Garrett came

to Princeton for frequent visits. After Johnny died, Garrett wrote a memoir about the work that Johnny did in the 1930s. Here is a sentence from Garrett's memoir: "Anyone wishing to get an unforgettable impression of the razor edge of von Neumann's mind need merely try to pursue this chain of exact reasoning for himself, realizing that often five pages of it were written down before breakfast, seated at a living room writing-table in a bathrobe."

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**9.** J. von Neumann, *Mathematische Grundlagen der Quantenmechanik,* Springer, Berlin, 1932.

**10.** ———, "Proof of the quasi-ergodic hypothesis," Proc. Nat. Acad. Sci. **18**, 70-82 (1932).

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# MATHEMATICAL FOUNDATIONS OF QUANTUM MECHANICS

# INTRODUCTION

The object of this book is to present the new quantum mechanics in a unified representation which, so far as it is possible and useful, is mathematically rigorous. This new quantum mechanics has in recent years achieved in its essential parts what is presumably a definitive form: the so-called "transformation theory." Therefore the principal emphasis shall be placed on the general and fundamental questions which have arisen in connection with this theory. In particular, the difficult problems of interpretation, many of which are even now not fully resolved, will be investigated in detail. In this connection, the relation of quantum mechanics to statistics and to the classical statistical mechanics is of special importance. However, we shall as a rule omit any discussion of the application of quantum mechanical methods to particular problems, as well as any discussion of special theories derived from the general theory—at least so far as this is possible without endangering the understanding of general relationships. This seems the more advisable since several excellent treatments of these problems are either in print or in process of publication.[1]

On the other hand, a presentation of the mathematical tools necessary for the purposes of this theory will be given, *i.e.*, a theory of Hilbert space and the so-called Hermitian operators. For this end, an accurate introduction to unbounded operators is also necessary, *i.e.*, an extension of the theory beyond its classical limits (developed by D. Hilbert and E. Hellinger, F. Riesz, E. Schmidt, O. Toeplitz). The following may be said regarding the method employed in this mode of treatment: as a rule, calculations should be performed with the operators themselves (which represent physical quantities) and not with the matrices, which, after the introduction of a (special and arbitrary) coordinate system in Hilbert space, result from them. This "coordinate free," *i.e.*, invariant, method, with its strongly geometrical language, possesses noticeable formal advantages.

[1] There are, among others, the following comprehensive treatments: Sommerfeld, Supplement to the 4th edition of *Atombau und Spectrallinien*, Braunschweig, 1928; Weyl, *The Theory of Groups and Quantum Mechanics* (translated by H. P. Robertson), London, 1931; Frenkel, *Wave Mechanics*, Oxford, 1932; Born and Jordan, *Elementare Quantenmechanik*, Berlin, 1930; Dirac, *The Principles of Quantum Mechanics*, 2nd edition, Oxford, 1936.

Dirac, in several papers, as well as in his recently published book,[2] has given a representation of quantum mechanics which is scarcely to be surpassed in brevity and elegance, and which is at the same time of invariant character. It is therefore perhaps fitting to advance a few arguments on behalf of our method, which deviates considerably from that of Dirac.

The method of Dirac, mentioned above (and this is overlooked today in a great part of the quantum mechanical literature, because of the clarity and elegance of the theory) in no way satisfies the requirements of mathematical rigor—not even if these are reduced in a natural and proper fashion to the extent common elsewhere in theoretical physics. For example, the method adheres to the fiction that every self-adjoint operator can be put in diagonal form. In the case of those operators for which this is not actually the case, this requires the introduction of "improper" functions with self-contradictory properties. The insertion of such a mathematical "fiction" is frequently necessary in Dirac's approach, even though the problem at hand is merely one of calculating numerically the result of a clearly defined experiment. There would be no objection here if these concepts, which cannot be incorporated into the present-day framework of analysis, were intrinsically necessary for the physical theory. Thus, as Newtonian mechanics first brought about the development of the infinitesimal calculus, which, in its original form, was undoubtedly not self-consistent, so quantum mechanics might suggest a new structure for our "analysis of infinitely many variables"—*i.e.*, the mathematical technique would have to be changed, and not the physical theory. But this is by no means the case. It should rather be pointed out that the quantum mechanical "Transformation theory" can be established in a manner which is just as clear and unified, but which is also without mathematical objections. It should be emphasized that the correct structure need not consist in a mathematical refinement and explanation of the Dirac method, but rather that it requires a procedure differing from the very beginning, namely, the reliance on the Hilbert theory of operators.

In the analysis of fundamental questions, it will be shown how the statistical formulas of quantum mechanics can be derived from a few qualitative, basic assumptions. Furthermore, there will be a detailed discussion of the problem as to whether it is possible to trace the statistical character of quantum mechanics to an ambiguity (*i.e.*, incompleteness) in our description of nature. Indeed, such an interpretation would be a natural concomitant of the general principle that every probability statement arises from the incompleteness of our knowledge. This explanation "by hidden parameters," as well as another, related to it, which ascribes the "hidden parameter" to the observer and not to the observed system, has been proposed more than once. However, it will appear that this can scarcely succeed in a satisfactory way, or more precisely, such an explanation

[2] See Proc. Roy. Soc. London, **109** (1925) and subsequent issues, especially **113** (1926). Independently of Dirac, P. Jordan, (Z. Physik **40** (1926)) and F. London (Z. Physik **40** (1926)) gave similar foundations for the theory.

is incompatible with certain qualitative fundamental postulates of quantum mechanics.[3]

The relation of quantum statistics to thermodynamics is also considered. A closer investigation shows that the well-known difficulties of classical mechanics, which are related to the "disorder" assumptions necessary for the foundations of thermodynamics, can be eliminated here.[4]

[3] See **IV** and **VI.3**

[4] See **V**.

# CHAPTER I

# INTRODUCTORY CONSIDERATIONS

## 1. THE ORIGIN OF THE TRANSFORMATION THEORY

This is not the place to point out the great success which the quantum theory attained in the period from 1900 to 1925, a development which was dominated by the names of Planck, Einstein and Bohr.[5] At the end of this period of development, it was clear beyond doubt that all elementary processes, *i.e.,* all occurrences of an atomic or molecular order of magnitude, obey the "discontinuous" laws of quanta. There were also available quantitative, quantum theoretical methods in almost all directions, which for the most part yielded results in good or at least fair agreement with experiment. And, what was fundamentally of greater significance, was that the general opinion in theoretical physics had accepted the idea that the principle of continuity ("*natura non facit saltus*"), prevailing in the macroscopic world, is merely simulated by an averaging process in a world which in truth is discontinuous by its very nature. This simulation is such that a man generally perceives the sum of many billions of elementary processes simultaneously, so that the leveling law of large numbers completely obscures the real nature of the individual processes.

Nevertheless, up to the time mentioned there existed no mathematical-physical system of quantum theory which would have embodied everything known up to that time in a unified structure, let alone one which could have

[5] Its chief stages were: The discovery of the quantum laws by Planck for the case of black body radiation (see Planck's presentation in his book, *Theory of Heat Radiation* (translated by M. Masius), Philadelphia, 1914); the hypothesis of the corpuscular nature of light (theory of light quanta) by Einstein (Ann. Phys. **17** (1905)), wherein the first example was given of the dual form: wave-corpuscle, which, we know today, dominates all of microscopic physics; the application of these two sets of rules to the atomic model by Bohr (Phil. Mag. **26** (1913); Z. Physik **6** (1920)).

exhibited the monumental solidity of the system of mechanics, electrodynamics and relativity theory (which system was disrupted by the quantum phenomena). In spite of the claim of quantum theory to universality, which had evidently been vindicated, there was lacking the necessary formal and conceptual instrument; there was a conglomeration of essentially different, independent, heterogeneous and partially contradictory fragments. The most striking points in this respect were: the correspondence principle, belonging half to classical mechanics and electrodynamics (but which played a decisive role in the final clarification of the problem); the self-contradictory dual nature of light (wave and corpuscular, cf. Note 5 and Note 148); and finally, the existence of unquantized (aperiodic) and quantized (periodic or multiply periodic) motions.[6]

The year 1925 brought the resolution. A procedure initiated by Heisenberg was developed by Born, Heisenberg, Jordan, and a little later by Dirac, into a new system of quantum theory, the first complete system of quantum theory which physics has possessed. A little later Schrödinger developed the "wave mechanics" from an entirely different starting point. This accomplished the same ends, and soon proved to be equivalent to the Heisenberg, Born, Jordan and Dirac system (at least in a mathematical sense, cf. **3** & **4** below).[7] On the basis of the Born statistical interpretation of the quantum theoretical description[8] of nature, it was possible for Dirac and Jordan[9] to join the two theories into one, the "transformation theory," in which they make possible a grasp of physical problems which is especially simple mathematically.

It should be mentioned (although it does not belong to our particular subject) that after Goudsmit and Uhlenbeck had discovered the magnetic moment and the spin of the electron, almost all the difficulties of the earlier quantum theory disappeared, so that today we are in possession of a mechanical system which is almost entirely satisfactory. To be sure, the great unity with electrodynamics and relativity theory mentioned earlier has not yet been recovered, but at least there is a mechanics which is universally valid, where the quantum laws fit in a natural and necessary manner, and which explains

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[6] The quantum laws (added to the laws of mechanics) for multiply periodic motions were first developed by Epstein-Sommerfeld [see, for example, Sommerfeld, *Atombau und Spectrallinien*, Braunscheig (1924)]. On the other hand, it was ascertained that a freely moving mass point of a planet on an hyperbolic orbit (in contrast to those on elliptical orbits) is "unquantized." The reader will find a complete treatment of this phase of quantum theory in the books by Reiche, *The Quantum Theory* (translated by H. S. Hatfield & H. L. Brose), New York, 1922; and Landé, *Fortschritte der Quantentheorie*, Dresden, 1922.

[7] This was proved by Schrödinger, Ann Physik **79** (1926).

[8] Z. Physik **37** (1926).

[9] See the articles mentioned in Note 2. Schrödinger's papers have been published in book form, *Collected Papers on Wave Mechanics* (translated by J. F. Shearer & W. M. Deans), London, 1928.

satisfactorily the majority of our experiments.[10]

## 2. THE ORIGINAL FORMULATIONS OF QUANTUM MECHANICS

In order to obtain a preliminary view of the problem, let us set forth briefly the basic structure of the Heisenberg-Born-Jordan "matrix mechanics" and the Schrödinger "wave mechanics."

In both theories, a classical mechanical problem is initially proposed, which is characterized by a Hamiltonian function $H(q_1, \ldots, q_k, p_1, \ldots, p_k)$. (This means the following, as may be found in greater detail in textbooks of mechanics: Let the system have $k$ degrees of freedom, *i.e.,* let its existing state be determined by giving that numerical values of $k$ coordinates $q_1, \ldots, q_k$. The energy is a given function of the coordinates and their time derivatives:

$$E = L(q_1, \ldots, q_k, \dot{q}_1, \ldots, \dot{q}_k)$$

and, as a rule, is a quadratic function of the derivatives $\dot{q}_1, \ldots, \dot{q}_k$. The "conjugate momenta" $p_1, \ldots, p_k$ of the coordinates $q_1, \ldots, q_k$ are introduced by the relations

$$p_1 = \frac{\partial L}{\partial \dot{q}_1}, \ \ldots, \ p_k = \frac{\partial L}{\partial \dot{q}_k}$$

In the case of the above assumption on $L$, these depend linearly on the $q_1, \ldots, q_k$. If need be, we can eliminate the $\dot{q}_1, \ldots, \dot{q}_k$ from $L$ by use of the $p_1, \ldots, p_k$, so that

$$E = L(q_1, \ldots, q_k, \dot{q}_1, \ldots, \dot{q}_k) = H(q_1, \ldots, q_k, p_1, \ldots, p_k)$$

(This $H$ is the Hamiltonian function.) In both theories, we must now learn as much as possible from this Hamiltonian function about the true, *i.e.,* quantum mechanical, behavior of the system. Primarily, therefore, we must determine[11]

[10] The present state of affairs may be described this way, that the theory, so far as it deals with individual electrons or with electronic shells of atoms or molecules, is entirely successful, as it is also whenever it deals with electrostatic forces and with electromagnetic processes connected with the production, transmission and transformation of light. On the other hand, in problems of the atomic nucleus, and in all attempts to develop a general and relativistic theory of electromagnetism, in spite of noteworthy partial successes, the theory seems to lead to great difficulties, which apparently cannot be overcome without the introduction of wholly new ideas.

[11] Motion, according to classical mechanics, is determined (as is well known) by the Hamiltonian function, since it gives rise to the equations of motion

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i} \quad : \quad i = 1, 2, \ldots, k$$

Before the discovery of quantum mechanics, the attempt was made to explain the quantum phenomena, while retaining these equations of motion, by the

the possible energy levels, then find out the corresponding "stationary states," and calculate the "transition probabilities," etc.[12]

The directions which the matrix theory gives for the solution of this problem run as follows: We seek a system of $2k$ matrices $\mathbb{Q}_1, \ldots, \mathbb{Q}_k, \mathbb{P}_1, \ldots, \mathbb{P}_k$,[13] which in the first place satisfy the relations

$$\left.\begin{aligned} \mathbb{Q}_m\mathbb{Q}_n - \mathbb{Q}_n\mathbb{Q}_m &= \mathbb{O} \\ \mathbb{P}_m\,\mathbb{P}_n - \mathbb{P}_n\,\mathbb{P}_m &= \mathbb{O} \\ \mathbb{P}_m\,\mathbb{Q}_n - \mathbb{Q}_n\,\mathbb{P}_m &= \delta_{mn}\tfrac{h}{2\pi i}\mathbb{I} \end{aligned}\right\} \quad (m, n = 1, 2, \ldots, k)$$

and for which, in the second place, the matrix

$$\mathbb{W} = H(\mathbb{Q}_1, \ldots, \mathbb{Q}_k, \mathbb{P}_1, \ldots, \mathbb{P}_k)$$

becomes a diagonal matrix. (We shall not go into the details here of the origin of these equations, especially the first group, the so-called "commutation rules" which govern the whole non-commutative matrix calculus of this theory. The reader will find exhaustive treatments of this subject in the works cited in Note 1. The quantity $h$ $[= 2\pi\hbar]$ is Planck's constant. The diagonal elements of $\mathbb{W}$, say

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formulation of supplementary quantum conditions (see Note 6). For each set of values of $q_1, \ldots, q_k, p_1, \ldots, p_k$, given at time $t = 0$, the equations of motion determined the further time variation, or "orbit," of the system in the $2k$ dimensional "phase space" $q_1, \ldots, q_k, p_1, \ldots, p_k$. Any additional condition, therefore, results in a limitation of all possible initial values or orbits to a certain discrete set. (Then, corresponding to the few admissible orbits, there is only a smaller number of possible energy levels.) Even though quantum mechanics has broken completely with this method, it is nevertheless clear from the outset that the Hamiltonian function must still play a great role in it. Indeed, common experience proves the validity of the Bohr correspondence principle, which asserts that the quantum theory must give results in agreement with those of classical mechanics in the so-called limiting case of large quantum numbers.

[12] The three latter concepts are taken from the old quantum theory developed principally by N. Bohr. Later we shall analyze these ideas in detail from the point of view of quantum mechanics. See the Dirac theory of radiation given in **III.6**. Their historical development can be followed in Bohr's papers on the structure of the atom, published from 1913 to 1916.

[13] As a more detailed analysis would show, this is a problem of infinite matrices. We shall not go farther here into the properties of such matrices, since we shall consider them thoroughly later on. For the moment it suffices that the formal algebraic calculation with these matrices is to be understood in the sense of the known rules of matrix addition and multiplication. By $\mathbb{O}$ and $\mathbb{I}$ we mean the null matrix and the unit matrix respectively (with all elements identically zero, and with elements equal to 1 on the main diagonal and zero everywhere else, respectively).

$w_1, w_2, \ldots$ are then the different allowed energy levels of the system. The elements of the matrices $\mathbb{Q}_1, \ldots, \mathbb{Q}_k$—$q^{(1)}_{mn}, \ldots, q^{(k)}_{mn}$—determine in a certain way the transition probabilities of the system (from the $m^{\text{th}}$ state with energy $w_m$ to the $n^{\text{th}}$ state with energy $w_n$, $w_m > w_n$) and the radiation thereby emitted.

In addition, it should be noted that the matrix

$$\mathbb{W} = H(\mathbb{Q}_1, \ldots, \mathbb{Q}_k, \mathbb{P}_1, \ldots, \mathbb{P}_k)$$

is not completely determined by the $\mathbb{Q}_1, \ldots, \mathbb{Q}_k, \mathbb{P}_1, \ldots, \mathbb{P}_k$ and the classical Hamiltonian function

$$H(q_1, \ldots, q_k, p_1, \ldots, p_k)$$

inasmuch as the $\mathbb{Q}_i$ and $\mathbb{P}_i$ do not all commute with one another (in multiplication), while it would be meaningless to distinguish between say $p_1 q_1$ and $q_1 p_1$ for $H(q_1, \ldots, q_k, p_1, \ldots, p_k)$ in the classical mechanical sense. We must therefore determine the order of the variables $q_i$ and $p_i$ in the terms of $H$, beyond the classical meaning of this expression. This process has not been carried out with complete generality, but the appropriate arrangements are known for the most important special cases. (In the simplest case, whenever the system under investigation consists of particles, and therefore has $k = 3\nu$ coordinates $q_1, \ldots, q_{3\nu}$—such that, e.g., $q_{3\mu-2}, q_{3\mu-1}, q_{3\mu}$ are the three cartesian coordinates of the $\mu^{\text{th}}$ particle, $\mu = 1, \ldots, \nu$—in which the interaction of these particles is given by a potential energy $V(q_1, \ldots, q_{3\nu})$, there is no such doubt. The classical Hamiltonian is then

$$H(q_1, \ldots, q_{3\nu}, p_1, \ldots, p_{3\nu}) = \sum_{\mu=1}^{\nu} \frac{1}{2m_\mu}(p^2_{3\mu-2} + p^2_{3\mu-1} + p^2_{3\mu}) + V(q_1, \ldots, q_{3\nu})$$

where $m_\mu$ is the mass of the $\mu^{\text{th}}$ particle, and $p_{3\mu-2}, p_{3\mu-1}, p_{3\mu}$ are the components of its momentum. What this means after the substitution of the matrices

$$\mathbb{Q}_1, \ldots, \mathbb{Q}_{3\nu}, \mathbb{P}_1, \ldots, \mathbb{P}_{3\nu}$$

is perfectly clear; in particular, the potential energy introduces no difficulties, since all the $\mathbb{Q}_1, \ldots, \mathbb{Q}_{3\nu}$ commute with each other.) It is important that only Hermitian matrices be permitted, *i.e.*, such matrices $\mathbb{A} = \{a_{mn}\}$, for which $a_{mn} = \overline{a_{nm}}$ holds identically (the elements $a_{mn}$ may be complex!). Therefore

$$H(\mathbb{Q}_1, \ldots, \mathbb{Q}_k, \mathbb{P}_1, \ldots, \mathbb{P}_k)$$

must be Hermitian, whenever all the $\mathbb{Q}_1, \ldots, \mathbb{Q}_k, \mathbb{P}_1, \ldots, \mathbb{P}_k$ are such. This involves a certain restriction in the problem of the order of the factors which was mentioned above. However, the restriction is not sufficient to determine the

$H(\mathbb{Q}_1,\dots,\mathbb{Q}_k,\mathbb{P}_1,\dots,\mathbb{P}_k)$ uniquely from the classical $H(q_1,\dots,q_k,p_1,\dots,p_k)$.[14]

On the other hand, the directions of wave mechanics are the following: First we form the Hamiltonian $H(q_1,\dots,q_k,p_1,\dots,p_k)$ and then the differential equation

$$H(q_1,\dots,q_k,\tfrac{\hbar}{i}\tfrac{\partial}{\partial q_1},\dots,\tfrac{\hbar}{i}\tfrac{\partial}{\partial q_k})\psi(q_1,\dots,q_k)=\lambda\psi(q_1,\dots,q_k)$$

for an arbitrary function $\psi(q_1,\dots,q_k)$ in the configuration space of the system (and not the phase space, *i.e.*, the $p_1,\dots,p_k$ do not enter into $\psi$). In this way,

$$H(q_1,\dots,q_k,\tfrac{\hbar}{i}\tfrac{\partial}{\partial q_1},\dots,\tfrac{\hbar}{i}\tfrac{\partial}{\partial q_k})$$

is interpreted simply as a functional operator. For example, this operator in the case mentioned above,

$$H(q_1,\dots,q_{3\nu},p_1,\dots,p_{3\nu})=\sum_{\mu=1}^{\nu}\frac{1}{2m_\mu}(p_{3\mu-2}^2+p_{3\mu-1}^2+p_{3\mu}^2)+V(q_1,\dots,q_{3\nu})$$

transforms the function $\psi(q_1,\dots,q_k)$ into

$$\sum_{\mu=1}^{\nu}\frac{1}{2m_\mu}\Big(\frac{\hbar}{i}\Big)^2\Big(\frac{\partial^2}{\partial q_{3\mu-2}^2}\psi+\frac{\partial^2}{\partial q_{3\mu-1}^2}\psi+\frac{\partial^2}{\partial q_{3\mu}^2}\psi\Big)+V\psi$$

(we have omitted the variables $q_1,\dots,q_{3\nu}$). Since the operation

$$q_1\tfrac{\hbar}{i}\tfrac{\partial}{\partial q_1}$$

is different from the operation[15]

$$\tfrac{\hbar}{i}\tfrac{\partial}{\partial q_1}q_1$$

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[14] If $\mathbb{Q}_1,\mathbb{P}_1$ are Hermitian, neither $\mathbb{Q}_1\mathbb{P}_1$ nor $\mathbb{P}_1\mathbb{Q}_1$ is necessarily Hermitian, but it is true that $\frac{1}{2}(\mathbb{Q}_1\mathbb{P}_1+\mathbb{P}_1\mathbb{Q}_1)$ is always Hermitian. In the case of $\mathbb{Q}_1^2\mathbb{P}_1$, we should also consider $\frac{1}{2}(\mathbb{Q}_1^2\mathbb{P}_1+\mathbb{P}_1\mathbb{Q}_1^2)$ as well as $\mathbb{Q}_1\mathbb{P}_1\mathbb{Q}_1$ (however, these two expressions happen to be equal for $\mathbb{P}_1\mathbb{Q}_1-\mathbb{Q}_1\mathbb{P}_1=(\hbar/i)\mathbb{I}$). In the case $\mathbb{Q}_1^2\mathbb{P}_1^2$, we should also consider $\frac{1}{2}(\mathbb{Q}_1^2\mathbb{P}_1^2+\mathbb{P}_1^2\mathbb{Q}_1^2)$, $\mathbb{Q}_1\mathbb{P}_1^2\mathbb{Q}_1$, $\mathbb{P}_1\mathbb{Q}_1^2\mathbb{P}_1$, etc. (these expressions do not all coincide in the special case mentioned above). We shall not discuss this further at present, since the operator calculus developed later will permit these relations to be seen much more clearly.

[15] We have

$$\tfrac{\hbar}{i}\tfrac{\partial}{\partial q_1}(q_1\psi)=q_1\tfrac{\hbar}{i}\tfrac{\partial}{\partial q_1}\psi+\tfrac{\hbar}{i}\psi$$

Consequently

$$\tfrac{\hbar}{i}\tfrac{\partial}{\partial q_1}\cdot q_1-q_1\cdot\tfrac{\hbar}{i}\tfrac{\partial}{\partial q_1}=\tfrac{\hbar}{i}\mathsf{I}$$

where $\mathsf{I}$ is the identity operator (transforming $\psi$ into itself), *i.e.*, $\frac{\hbar}{i}\frac{\partial}{\partial q_1}$ and $q_1$ satisfy the same commutation rules as the matrices $\mathbb{P}_1$ and $\mathbb{Q}_1$.

there is here, too, an uncertainty because of the ambiguity of the order of the terms $q_i$ and $p_i$ in

$$H(q_1, \ldots, q_k, p_1, \ldots, p_k)$$

However, Schrödinger has pointed out how this uncertainty can be eliminated, by reduction to a definite variational principle, in such a way that the resulting differential equation becomes self-adjoint.[16]

Now this differential equation (the "wave equation") has the character of an eigenvalue problem in which $\lambda$ is to be interpreted as an eigenvalue parameter, and in which the vanishing of the eigenfunction $\psi = \psi(q_1, \ldots, q_k)$ at the boundaries of the configuration space (the space of the $q_1, \ldots, q_k$)—together with the conditions of regularity and single-valuedness—is required. In the sense of the wave theory, the eigenvalues of $\lambda$ (both discrete and continuous spectra)[17] are the allowed energy levels. And even the corresponding (complex) eigenfunctions are related to the corresponding (stationary, in the Bohr sense) states of the system. For a $\nu$-electron system ($k = 3\nu$, see above; $e$ is the charge of the electron) the charge density of the $\mu^{\text{th}}$ electron, measured at the point $x, y, x$, is given by the expression

$$e \underbrace{\int \cdots \int}_{(3\nu-3)\text{fold}} |\psi(q_1 \ldots q_{3\mu-3} x y z q_{3\mu+1} \ldots q_{3\nu})|^2 dq_1 \cdots dq_{3\mu-3} dq_{3\mu+1} \cdots dq_{3\nu}$$

*i.e.*, according to Schrödinger, this electron is to be thought of as "smeared" over the entire $x, y, z$ ($= q_{3\mu-2}, q_{3\mu-1}, q_{3\mu}$) space. (In order that the total charge be $e$, $\psi$ must be normalized by the condition

$$\underbrace{\int \cdots \int}_{(3\nu)\text{fold}} |\psi(q_1, \ldots, q_{3\nu})|^2 dq_1 \cdots dq_{3\nu} = 1$$

The integration is over all $3\nu$ variables. The same equation obtains for each $\mu = 1, \ldots, \nu$.)

In addition, the wave mechanics can also make observations on systems which are not in Bohr stationary states,[18] in the following way: If the state is not stationary, *i.e.*, if it changes with time, then the wave function

$$\psi(q_1, \ldots, q_k; t)$$

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[16] See his first two articles, in the book mentioned in Note 9 (also Ann. Phys. **79** (1926)).

[17] See the first of the works of Schrödinger mentioned in Note 16. A precise definition of the spectrum and its parts will be given later, in **II.6** to **II.9**.

[18] In the original framework of matrix mechanics (see our presentation above), such a general state concept, of which the stationary states are special cases, was not given. Only the stationary states, arranged according to the values of the energy, were the object of that theory.

contains time $t$, and it varies according to the differential equation[19]

$$-H(q_1,\ldots,q_k,\tfrac{\hbar}{i}\tfrac{\partial}{\partial q_1},\ldots,\tfrac{\hbar}{i}\tfrac{\partial}{\partial q_k})\psi(q_1,\ldots,q_k;t)=\tfrac{\hbar}{i}\tfrac{\partial}{\partial t}\psi(q_1,\ldots,q_k;t)$$

That is, $\psi$ can be given arbitrarily for $t=t_0$, and it is then determined uniquely for all $t$. Even the stationary $\psi$ are really time-dependent, as a comparison of the two Schrödinger differential equations will show, but the dependence on $t$ is given by

$$\psi(q_1,\ldots,q_k;t)=e^{-i\lambda t/\hbar}\cdot\psi(q_1,\ldots,q_k)$$

That is, $t$ appears only in a factor of absolute value independent of $q_1,\ldots,q_k$ (*i.e.*, constant in configuration space), so that, for example, the charge density distribution defined above does not change. (We shall suppose generally—and we shall find this confirmed later by more detailed considerations—that a factor of absolute value 1 and constant in configuration space is, in the case of $\psi$, essentially unobservable.)

Since the eigenfunctions of the first differential equation form a complete orthogonal set,[20] we can develop each $\psi(q_1,\ldots,q_k)$ in terms of this set of functions. If $\psi_1,\psi_2,\ldots$ are the eigenfunctions (all independent of time), and $\lambda_1,\lambda_2,\ldots$ are their respective eigenvalues, the development becomes[21]

$$\psi(q_1,\ldots,q_k)=\sum_{n=1}^{\infty}a_n\psi_n(q_1,\ldots,q_k)$$

If $\psi$ is still time-dependent, then $t$ is introduced in the coefficients $a_n$ (the eigenfunctions $\psi_1,\psi_2,\ldots$ on the other hand are to be understood both here and in everything which follows as independent of time). Therefore, if the

$$\psi=\psi(q_1,\ldots,q_k)$$

at hand is actually $\psi=\psi(q_1,\ldots,q_k;t_0)$, then it follows with regard to

$$\psi=\psi(q_1,\ldots,q_k;t)=\sum_{n=1}^{\infty}a_n(t)\psi_n$$

that

$$H\psi=\sum_{n=1}^{\infty}a_n(t)H\psi_n=\sum_{n=1}^{\infty}\lambda_n a_n(t)\psi_n$$

$$\tfrac{\hbar}{i}\tfrac{\partial}{\partial t}\psi=\sum_{n=1}^{\infty}\tfrac{\hbar}{i}\dot{a}_n(t)\psi_n$$

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[19] $H=H(q_1,\ldots,q_k,p_1,\ldots,p_k)$ may also contain the time $t$ explicitly. Naturally, there will then be in general no stationary states at all.

[20] Provided that only a discrete spectrum exists. See **II.6**.

[21] These, as also all the following series expansions, converge "in the mean." We shall consider this again in **II.2**.